

Toward Transformer-Based Neural Posterior Estimation for Bayesian Dynamic Borrowing with Longitudinal Alzheimer’s Disease Data

Anonymous Author(s)

Anonymous Institution

ANONYMOUS@ANONYMOUS.EDU

Abstract

Leveraging external or historical cohorts to augment small concurrent studies is attractive in Alzheimer’s disease research, where longitudinal follow-up is expensive and disease trajectories are heterogeneous. Bayesian dynamic borrowing (BDB) provides a principled mechanism for adaptively controlling the influence of external data, but classical implementations rely on low-dimensional parametric summaries that fail to capture the richness of longitudinal clinical sequences. We propose combining transformer-based patient-trajectory embeddings with amortized neural posterior estimation (NPE) to address this limitation. The transformer encoder learns high-quality representations from irregular event sequences in Alzheimer’s Disease Neuroimaging Initiative (ADNI) records, encoding clinical concepts, domain type, and temporal structure jointly. The NPE component then maps source-level embeddings and discrepancy features to a posterior distribution over the current-study estimand, amortizing inference over many simulated borrowing configurations and eliminating the need for repeated MCMC computation at test time. This paper establishes the background, related work, and problem formulation for this framework. We situate the approach within the literature on BDB, balancing methods, simulation-based inference, and transformer models for electronic health records. We formalize the inference target under five exchangeability regimes (SC1–SC5) covering full exchangeability, partial and strong covariate shift, and partial and strong outcome drift. We also outline an embedding-visualization strategy using t-SNE or UMAP to evaluate representation quality before and after transformer encoding, and describe the planned empirical workflow with ADNI as the downstream validation dataset.

1. Introduction

Alzheimer’s disease (AD) and related dementias are a growing public health problem, with limited options for disease-modifying treatment and substantial heterogeneity in how the disease progresses across individuals. Longitudinal studies such as the Alzheimer’s Disease Neuroimaging Initiative (ADNI) ([Petersen et al., 2010](#); [Weiner et al., 2017](#)) have collected multi-modal data over many years per participant, including cognitive assessments, cerebrospinal fluid and imaging biomarkers, medication records, and clinical events. One natural use of such data is to serve as an external control arm for a smaller concurrent study—for example, using participants from an earlier ADNI phase to augment a new target cohort. The problem is that data from different ADNI phases, or from any two real-world cohorts, are often not directly comparable. Differences in enrollment criteria, assessment protocols, standard-of-care practices, or disease stage mix can introduce substantial bias if external data is borrowed without adjustment ([Viele et al., 2014](#)).

Bayesian dynamic borrowing (BDB) provides a principled way to handle this by letting the amount of external information incorporated adapt to how consistent the two datasets appear ([Ibrahim](#)

and Chen, 2000; Hobbs et al., 2011; Neuenschwander et al., 2010; Schmidli et al., 2014). Methods like the power prior, commensurate prior, and robust MAP prior have been used in pharmaceutical trials and have clear statistical interpretations. Their main limitation in a longitudinal setting is that they reduce source-level disagreement to one or a few scalar parameters. In ADNI data, two cohorts might look similar on baseline demographics but differ substantially in visit frequency, cognitive decline rates, or the sequence in which biomarkers become abnormal—none of which a single discrepancy number can capture well. There is also a practical issue: fitting these models from scratch on every new pair of cohorts via MCMC is slow, which becomes a real bottleneck when screening many candidate external sources.

Amortized neural posterior estimation (NPE) addresses the computational side of this problem (Papamakarios et al., 2019; Greenberg et al., 2019; Cranmer et al., 2020; Lückmann et al., 2021). The idea is to train a neural network on many simulated dataset pairs, so that at test time it can output a posterior over the treatment effect with a single forward pass rather than running MCMC again. For this to work well, the input to the network needs to carry enough information about both the treatment effect and the degree of cohort mismatch. Simple summary statistics like group means and variances miss the temporal structure of longitudinal trajectories, which is often exactly where the relevant differences lie.

We therefore propose using a transformer encoder to represent patient trajectories before passing cohort summaries into the NPE network. Transformers have been applied successfully to EHR sequences in models like BEHRT (Li et al., 2020) and Med-BERT (Rasmy et al., 2021), and their self-attention mechanism handles the irregular timing and variable length that characterize ADNI visits naturally. Beyond predicting outcomes within a single cohort, the encoder here must produce representations that make source differences visible—so the posterior network can tell from the embeddings whether the external cohort is close enough to borrow from.

The rest of this paper covers background and problem setup. Section 2 reviews BDB methods, weighting approaches, and simulation-based inference. Section 3 defines the inference target and the five exchangeability regimes (SC1–SC5) used to evaluate the framework. Section 4 sketches figure templates and the planned empirical workflow with ADNI as the validation dataset.

Generalizable Insights about Machine Learning in the Context of Healthcare

- **Amortized inference removes a practical barrier to dynamic borrowing.** Replacing per-dataset MCMC with a pre-trained NPE network brings posterior computation down from minutes to milliseconds. This makes it realistic to screen many candidate external cohorts in real-world evidence studies, which was previously impractical.
- **Trajectory-level embeddings can capture non-exchangeability that scalar diagnostics miss.** Two AD cohorts may look similar on baseline demographics but diverge in temporal event patterns or progression rates. A transformer that encodes these sequences can expose such differences before any posterior inference is done, giving the network a better signal for how much to borrow.
- **A five-regime simulation design provides a reusable evaluation benchmark.** The SC1–SC5 framework—covering full exchangeability through strong outcome drift—gives a structured way to stress-test borrowing methods that applies to any longitudinal study with a candidate external cohort, not just ADNI.

2. Related Work

2.1. Bayesian dynamic borrowing

Bayesian dynamic borrowing aims to combine a current dataset D_c with an external dataset D_e while limiting the damage when the two sources are not fully compatible. The fundamental trade-off is well understood: borrowing more reduces variance but risks bias when the sources differ; borrowing less is conservative but wastes available data. The methods below operationalize this trade-off in different ways.

Power prior. The power prior (Ibrahim and Chen, 2000; Ibrahim et al., 2015) down-weights the external likelihood by raising it to a power $a_0 \in [0, 1]$:

$$p(\theta \mid D_c, D_e, a_0) \propto L(\theta; D_c) L(\theta; D_e)^{a_0} \pi(\theta). \quad (1)$$

When $a_0 = 1$ the data are fully pooled; $a_0 = 0$ ignores the external data. The normalized power prior (Duan et al., 2006) treats a_0 as a random variable and integrates over it, so the amount of borrowing is determined by the data rather than fixed in advance.

Commensurate prior. Hobbs et al. (2011) use separate parameters for each source and link them through a precision parameter τ :

$$\theta_e \sim \pi(\theta_e), \quad (2)$$

$$\theta_c \mid \theta_e, \tau \sim \mathcal{N}(\theta_e, \tau^{-1}). \quad (3)$$

Small τ allows the two effects to differ freely; large τ forces them together. Placing a heavy-tailed prior on τ lets the model reduce borrowing when the data suggest conflict.

Robust MAP prior. The robust meta-analytic-predictive prior (Neuenschwander et al., 2010; Schmidli et al., 2014) mixes an informative prior derived from historical data with a vague component:

$$\pi_{\text{robust}}(\theta_c) = (1 - \epsilon) \pi_{\text{MAP}}(\theta_c) + \epsilon \pi_{\text{vague}}(\theta_c). \quad (4)$$

The weight ϵ is pre-specified and controls how much protection against prior-data conflict is built in. This approach is commonly used in pharmaceutical regulatory submissions because the mixing weight has a clear, auditable interpretation.

Limitations in longitudinal settings. All three approaches summarize source agreement with one or a few numbers. In ADNI-style longitudinal data, disagreement between two cohorts can show up simultaneously in visit intensity, biomarker trajectories, cognitive test profiles, medication patterns, and outcome ascertainment. Squeezing all of that into a scalar discrepancy parameter loses too much information. The approach developed here uses learned trajectory embeddings instead, which can reflect multiple dimensions of source mismatch at once.

2.2. Balancing and weighting approaches

A separate literature addresses source differences through reweighting rather than hierarchical priors. Propensity-score methods reweight external subjects so their covariate distribution matches the current cohort (Stuart, 2010). Inverse-probability weighting (IPW) is the standard implementation, though it can be unstable when propensity scores are near zero or one. Overlap weighting (Li et al.,

2018) sidesteps this by giving more weight to subjects with propensity scores near 0.5, i.e., those in the region where the two cohorts actually overlap, using weights proportional to $2\hat{e}(x)(1 - \hat{e}(x))$.

The limitation shared by all weighting methods is that they correct for differences in who is in each cohort, but not for differences in how the outcome is generated (Viele et al., 2014; Dahabreh et al., 2019). In an AD study, the outcome mechanism can differ across enrollment periods due to changes in diagnostic criteria, availability of new medications, or differences in how cognitive decline is assessed—none of which reweighting on baseline covariates will fix. The proposed framework therefore includes both covariate-based discrepancy features and outcome-related diagnostics as inputs to the posterior network.

2.3. Neural posterior estimation and simulation-based inference

Neural posterior estimation (NPE) is a form of simulation-based inference (Cranmer et al., 2020) in which a neural network learns to map from observed data to an approximate posterior:

$$q_\phi(\theta | x) \approx p(\theta | x). \quad (5)$$

Training minimizes the expected negative log-likelihood over simulated pairs $(\theta, x) \sim p(\theta)p(x | \theta)$:

$$\mathcal{L}(\phi) = \mathbb{E}[-\log q_\phi(\theta | x)]. \quad (6)$$

At test time, the trained network produces a posterior in a single forward pass, eliminating the need to rerun MCMC (Papamakarios et al., 2019; Greenberg et al., 2019; Lückmann et al., 2021). Prior-fitted networks (PFNs) (Müller et al., 2022) take this further by using a transformer as the posterior network itself, conditioning directly on the dataset in-context.

Applying NPE to dynamic borrowing requires the network input x to encode both the treatment effect signal and the degree of cohort mismatch. Most existing NPE work uses fixed, hand-crafted summaries designed for scientific simulators with relatively clean structure. A preliminary simulation study using a stylized Gaussian data-generating mechanism and an OMOP-like longitudinal simulator has shown that adding covariate-shift diagnostics such as KL divergence to the summary vector reduces posterior bias under covariate mismatch (SC2, SC4), which supports the case for richer trajectory-level representations in the ADNI setting.

2.4. Transformers and patient trajectory embeddings

Transformers (Vaswani et al., 2017) are now widely used for EHR sequence modeling. BEHRT (Li et al., 2020) and Med-BERT (Rasmy et al., 2021) treat diagnosis codes, medications, and procedures as tokens and use BERT-style pretraining to learn patient representations, handling irregular visit timing through temporal position encodings.

The difference from standard EHR prediction is that the encoder here is not trained to predict a single outcome label. Instead, the representation needs to be useful for posterior inference under borrowing—it must summarize disease-relevant longitudinal structure within each cohort while also making any differences between cohorts apparent. Whether the encoder is achieving this can be checked with dimensionality reduction tools like t-SNE (van der Maaten and Hinton, 2008) or UMAP (McInnes et al., 2018): embeddings from exchangeable cohorts should overlap, while embeddings from non-exchangeable cohorts should separate.

3. Background and Problem Setup

3.1. Alzheimer’s disease data and ADNI

ADNI is a multi-site longitudinal study that has followed participants across the spectrum from cognitively normal to Alzheimer’s dementia (Petersen et al., 2010; Weiner et al., 2017). Participants are assessed at irregular intervals using cognitive batteries (ADAS-Cog, MMSE, CDR), cerebrospinal fluid and blood biomarkers (amyloid- β , phospho-tau), MRI and PET imaging, and clinical event records. Four study phases—ADNI-1, ADNI-GO, ADNI-2, and ADNI-3—differ in their enrollment criteria, sample sizes, and assessment protocols. These phase differences make ADNI a natural testbed for dynamic borrowing: one phase can serve as an external cohort for another, but the two may not be directly comparable because of protocol changes or shifts in enrollment criteria over time.

3.2. Inference target and borrowing setup

Let $D_c = \{(x_{ci}, y_{ci})\}_{i=1}^{n_c}$ be the current (target) dataset and $D_e = \{(x_{ej}, y_{ej})\}_{j=1}^{n_e}$ be the external dataset. For patient i in source $s \in \{c, e\}$, the observed data is a sequence of clinical visits:

$$X_{si} = \{(v_{sit}, t_{sit}), \dots, (v_{siT_{si}}, t_{siT_{si}})\},$$

where v_{sit} is the clinical content at visit time t_{sit} , which may include diagnosis codes, cognitive scores, laboratory values, medication records, or imaging summaries.

The inferential target is θ , a current-study estimand such as a treatment effect or progression contrast. The goal is to compute

$$p(\theta \mid D_c, D_e), \quad (7)$$

where the amount of information contributed by D_e should reflect how exchangeable the two cohorts are. Importantly, θ is defined for the current population; the framework does not require the external cohort to have the same underlying effect.

3.3. Exchangeability regimes (SC1–SC5)

We organize evaluation around five scenarios that represent distinct ways borrowing can succeed or fail:

Scenario	Interpretation
SC1	Fully exchangeable. The external and current cohorts have similar covariate distributions and the same outcome mechanism.
SC2	Partial covariate shift. Moderate differences in baseline characteristics or trajectory composition, but a similar outcome mechanism.
SC3	Partial outcome drift. Trajectory structure is similar across cohorts, but the outcome mechanism differs moderately.
SC4	Strong covariate shift. The external cohort differs substantially in case mix, assessment cadence, or temporal event patterns.
SC5	Strong outcome drift. Covariate distributions are similar, but the outcome mechanism differs materially—for example, because the cohorts span different disease stages or enrollment periods.

Weighting methods tend to work well for SC2 and SC4, where the problem is primarily covariate-driven. Adaptive Bayesian borrowing methods are designed for SC3 and SC5, where outcome drift is the main concern. The goal here is a method that handles all five without prior knowledge of which regime applies. A sixth scenario combining strong shift and strong drift simultaneously was considered but excluded from this first study, to keep the diagnosis of individual failure modes clean.

3.4. Why transformer-based embeddings?

Standard BDB summaries typically consist of group means, variances, and sample sizes. For tabular data, these capture most of what matters. For longitudinal data, they do not. Two ADNI cohorts might have nearly identical mean ages, sex ratios, and baseline cognitive scores yet differ substantially in how fast participants decline, how often they are seen, or what medications they receive over time. Those temporal differences can determine whether borrowing is safe, but they are invisible to cross-sectional summary statistics.

The approach here represents each patient trajectory using a transformer encoder. Each clinical event is embedded as

$$z_{sit}^{(0)} = E_{\text{concept}}(c_{sit}) + E_{\text{type}}(d_{sit}) + E_{\text{time}}(\Delta t_{sit}), \quad (8)$$

where c_{sit} is a clinical concept identifier (e.g., an OMOP concept code), d_{sit} is the clinical domain (condition, drug, measurement, or procedure), and Δt_{sit} is the time since the previous event. Self-attention layers produce a contextualized representation for each token, which are then pooled into a patient-level embedding $h_{si} \in \mathbb{R}^d$. Patient embeddings are aggregated within each source to form cohort-level summaries and discrepancy features that go into the posterior network.

A preliminary simulation study using an OMOP-like longitudinal data-generating mechanism showed that adding KL divergence between the covariate distributions of the two cohorts to the NPE input reduces bias in the SC2 and SC4 regimes. Transformer-derived embeddings carry that idea further, encoding distributional differences across the full temporal trajectory rather than just marginal covariate summaries.

3.5. Neural posterior estimation for borrowing

Given patient embeddings from both cohorts, source-level summaries $\bar{h}_c$ and $\bar{h}_e$ are formed by mean pooling or learned aggregation. Additional discrepancy features $\delta(D_c, D_e)$ may include KL divergence between embedding distributions, overlap statistics based on propensity scores, and outcome-related drift indicators. The full NPE input is

$$s(D_c, D_e) = [\bar{h}_c, \bar{h}_e, \delta(D_c, D_e)], \quad (9)$$

and the network learns

$$q_\phi(\theta | s(D_c, D_e)) \approx p(\theta | D_c, D_e), \quad (10)$$

where q_ϕ is a Gaussian or normalizing flow. Training draws simulated pairs (θ, D_c, D_e) from a simulator covering the SC1–SC5 regimes.

The design has a few practical advantages worth noting. Once training is done, inference on a new ADNI cohort pair requires only a forward pass—no MCMC. The transformer and the posterior network can also be trained jointly, so the encoder learns what aspects of trajectory differences actually matter for borrowing, not just what is predictive within a single cohort. Finally, the architecture

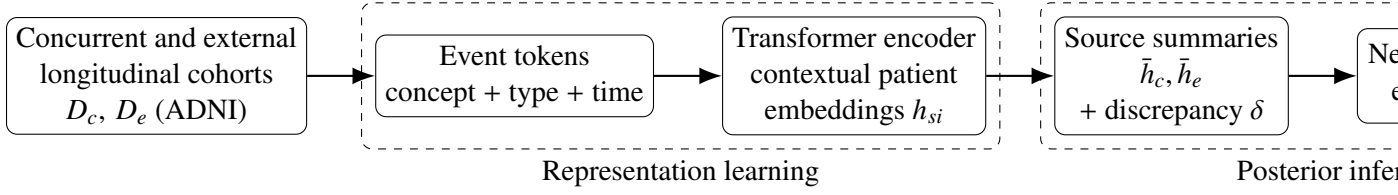


Figure 1: Planned transformer-based NPE workflow for adaptive dynamic borrowing with ADNI longitudinal data. The representation stage learns patient embeddings from irregular clinical trajectories; the posterior stage maps cohort-level summaries and discrepancy features to an approximate posterior over the current-study estimand θ .

is modular: the posterior network can initially take hand-crafted summaries as input and be upgraded to transformer embeddings once the ADNI pipeline is ready, without changing the inference logic.

3.6. Planned empirical workflow

The first empirical stage uses synthetic data. We evaluate the NPE estimator across SC1–SC5 and compare against commensurate priors, robust MAP priors, IPW, and overlap weighting. The simulation covers both the stylized Gaussian setting and the OMOP-like longitudinal simulator.

The second stage transfers to ADNI. One ADNI phase will serve as the external cohort and another as the target, with the estimand defined as a clinically meaningful progression or treatment-related contrast. Transformer embedding diagnostics—t-SNE and UMAP plots colored by phase and by SC label—will check whether the learned representations reflect the expected degree of source mismatch before any posterior results are reported.

4. Figures and Writing Plan

This section provides paper-ready figure templates that can be updated once the ADNI data pipeline and transformer training are complete.

4.1. Conceptual workflow figure

Figure 1 summarizes the planned modeling pipeline. Representation learning and posterior inference play distinct but coupled roles: the transformer produces high-quality longitudinal embeddings, and the neural posterior estimator converts source-level representations and discrepancy features into an approximate posterior for the current-study estimand.

4.2. Embedding visualization

A key diagnostic for the transformer stage is whether the learned embeddings separate clinically and statistically meaningful cohort configurations. We plan to present t-SNE (van der Maaten and Hinton, 2008) or UMAP (McInnes et al., 2018) visualizations in a two-by-two layout (before and after transformer encoding, colored by source and by scenario label). The target pattern is: embeddings

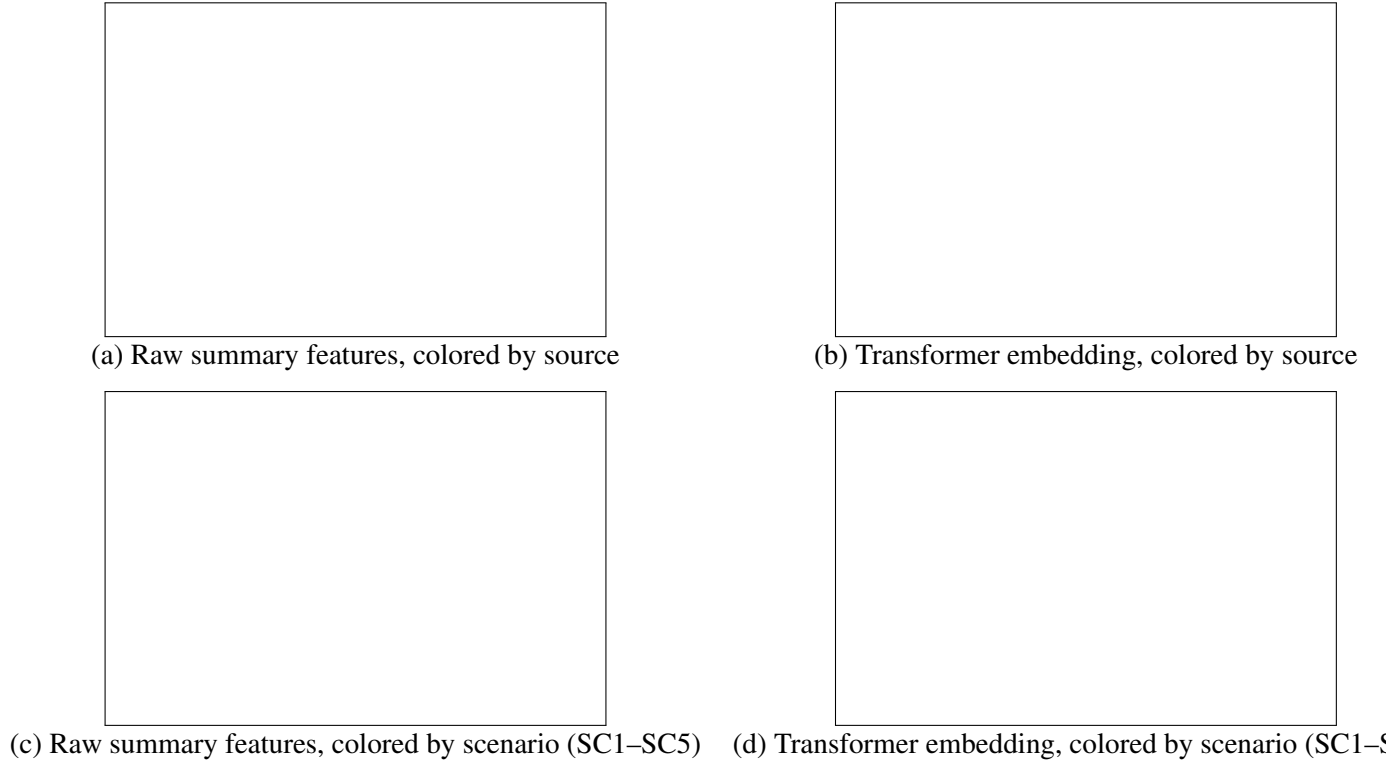


Figure 2: Placeholder for t-SNE/UMAP visualization. Replace each panel with the actual embedding plot once the ADNI representation pipeline is complete. (a, b) Color by source (current vs. external) to show overlap under SC1 and separation under SC4/SC5. (c, d) Color by scenario label to verify that the transformer separates exchangeability regimes more cleanly than raw summary features.

from exchangeable sources (SC1) should overlap, while embeddings from strongly mismatched sources (SC4, SC5) should separate.

4.3. Writing plan for the full manuscript

1. **Introduction** (done): motivation from AD research and ADNI, gap between classical BDB and longitudinal data richness, proposed transformer-NPE combination.
2. **Related work** (done): BDB methods (power prior, commensurate, robust MAP), balancing methods (IPW, overlap weighting), simulation-based inference and NPE, transformers for EHR.
3. **Problem setup** (done): ADNI data description, inference target, exchangeability regimes SC1–SC5, transformer embedding design, NPE formulation.
4. **Method section** (to write after ADNI task definition): transformer architecture details, training procedure, NPE network and posterior parameterization, joint or staged training strategy.

5. **Experiments:** simulation study under SC1–SC5 with synthetic data and OMOP-like simulator; ADNI case study with source partition and estimand definition; ablation of hand-crafted vs. transformer summaries; embedding diagnostics.
6. **Discussion:** conditions under which transformer embeddings improve borrowing, coverage calibration gaps, limitations, and future work on real-time ADNI deployment.

4.4. Planned baselines

The method comparison will include the following baselines:

- Commensurate prior ([Hobbs et al., 2011](#))
- Robust MAP prior ([Schmidli et al., 2014](#))
- Inverse-probability weighting (IPW) ([Stuart, 2010](#))
- Overlap weighting ([Li et al., 2018](#))

These methods span both Bayesian borrowing and balancing-based adjustment, providing a reference frame for evaluating whether transformer-informed NPE adds value beyond standard covariate adjustment.

Generative AI Use Disclosure

In preparing this manuscript, the authors used a large language model (Claude, Anthropic) to assist with prose editing, LaTeX formatting, and structural drafting of sections. All technical content, mathematical formulations, simulation designs, and scientific claims were developed, verified, and take full responsibility for by the authors. The LLM was not used to generate novel research ideas, produce data, or perform analysis.

References

- Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- Issa J. Dahabreh, Sarah E. Robertson, Eric J. Tchetgen, Elizabeth A. Stuart, and Miguel A. Hernan. Generalizing causal inferences from individuals in randomized trials to all trial-eligible individuals. *Biometrics*, 75(2):685–694, 2019.
- Yunzhang Duan, Keying Ye, and Eric P. Smith. A modified Bayesian framework for multi-source data with application to toxicology. *Biometrics*, 62(3):867–875, 2006.
- David S. Greenberg, Marcel Nonnenmacher, and Jakob H. Macke. Automatic posterior transformation for likelihood-free inference. *Proceedings of Machine Learning Research*, 97:2404–2414, 2019.
- Brian P. Hobbs, Bradley P. Carlin, Sumithra J. Mandrekar, and Daniel J. Sargent. Commensurate priors for incorporating historical information in clinical trials using general and generalized linear models. *Bayesian Analysis*, 6(4):639–674, 2011.

- Joseph G. Ibrahim and Ming-Hui Chen. Power prior distributions for regression models. *Statistical Science*, 15(1):46–60, 2000.
- Joseph G. Ibrahim, Ming-Hui Chen, Youngseok Gwon, and Fang Chen. The power prior: theory and applications. *Statistics in Medicine*, 34(28):3724–3749, 2015.
- Fan Li, Kari L. Morgan, and Alan M. Zaslavsky. Balancing covariates via propensity score weighting. *Journal of the American Statistical Association*, 113(521):390–400, 2018.
- Yikuan Li, Shishir Rao, J. R. A. Solares, Abdelrahman Hassaine, Rohan Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. Behrt: Transformer for electronic health records. *Scientific Reports*, 10(1):7155, 2020.
- Jan-Matthis Lückmann, Pedro J. Gonçalves, Giacomo Bassetto, Kaan Öcal, Marcel Nonnenmacher, and Jakob H. Macke. Benchmarking simulation-based inference. *Proceedings of Machine Learning Research*, 130:343–347, 2021.
- Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- Samuel Müller, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. Transformers can do Bayesian inference. In *International Conference on Learning Representations*, 2022.
- Beat Neuenschwander, Gaby Capkun-Niggli, Michael Branson, and David J. Spiegelhalter. Summarizing historical information on controls in clinical trials. *Clinical Trials*, 7(1):5–18, 2010.
- George Papamakarios, David Sterratt, and Iain Murray. Sequential neural likelihood: Fast likelihood-free inference with autoregressive flows. *Proceedings of Machine Learning Research*, 89:837–848, 2019.
- Ronald C. Petersen, Paul S. Aisen, Laurel A. Beckett, Michael C. Donohue, Anthony C. Gamst, Danielle J. Harvey, Clifford R. Jack, William J. Jagust, Leslie M. Shaw, Arthur W. Toga, John Q. Trojanowski, and Michael W. Weiner. Alzheimer’s disease neuroimaging initiative (adni): clinical characterization. *Neurology*, 74(3):201–209, 2010.
- Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. Med-bert: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *NPJ Digital Medicine*, 4(1):86, 2021.
- Heinz Schmidli, Sara Gsteiger, Shreerupa Roychoudhury, Anthony O’Hagan, David Spiegelhalter, and Beat Neuenschwander. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.
- Elizabeth A. Stuart. Matching methods for causal inference: A review and a look forward. *Statistical Science*, 25(1):1–21, 2010.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Kert Viele, Scott Berry, Beat Neuenschwander, Billy Amzal, Fang Chen, Nandini Enas, Brian Hobbs, Joseph G. Ibrahim, Nelson Kinnersley, Stacy Lindborg, et al. Use of historical control data for assessing treatment effects in clinical trials. *Pharmaceutical Statistics*, 13(1):41–54, 2014.
- Michael W. Weiner, Dallas P. Veitch, Paul S. Aisen, Laurel A. Beckett, Nigel J. Cairns, Robert C. Green, Danielle Harvey, Clifford R. Jack, William Jagust, Enchi Liu, John C. Morris, Ronald C. Petersen, Andrew J. Saykin, Leslie M. Shaw, Li Shen, Judith A. Siuciak, Holly Soares, Arthur W. Toga, and John Q. Trojanowski. The alzheimer’s disease neuroimaging initiative 3: continued innovation for clinical trial improvement. *Alzheimer’s & Dementia*, 13(5):561–571, 2017.